

direct Method:

$$(\bar{x}) = \frac{\sum fm}{\sum f} = \frac{4100}{100} = 41$$

Short Cut Method:

$$(\bar{x}) = A + \frac{\sum fd}{\sum f} = 45 - \frac{400}{100} = 41$$

Step Deviation Method:

$$(\bar{x}) = A + \frac{\sum fd'}{\sum f} \times C = 45 - \frac{40}{100} \times 10 = 41$$

Ex-15 Solv	x	f	m	d=m-A	C=10	f d'
					$d' = \frac{d}{C}$	
	0-10	50-45=5	5	-20	-2	-10
	10-20	45-38=7	15	-10	-1	-7
	20-30	38-26=12	25=A	0	0	0
	30-40	26-10=16	35	10	1	16
	40-50	10-4=6	45	20	2	12
	50-60	4	55	30	3	12
		$\sum f = 50$				$\sum fd' = 23$

$$(\bar{x}) = A + \frac{\sum fd'}{\sum f} \times C = 25 + \frac{23}{50} \times 10 = 29.6$$

Test Yourself 8.4

2-Solu

x	f	m	d=m-A	C=10	d' = $\frac{d}{C}$	f d'
140-150	25	145	-20		-2	-50
150-160	21	155	-10		-1	-21
160-170	15	165=A	0		0	0
170-180	7	175	10		1	7
180-190	2	185	20		2	2

$$\sum f = 70$$

$$\sum fd' = 62$$

$$(\bar{x}) = A + \frac{\sum fd'}{\sum f} \times C$$

$$= 165 - \frac{62}{70} \times 10$$

$$= 163 \text{ cm}$$

4-Solu

x	f	m	d=m-A	C=10	d' = $\frac{d}{C}$	f d'
21-30	5	24.5	-20		-2	-10
31-40	6	34.5	-10		-1	-6
41-50	7	A=44.5	0		0	0
51-60	8	54.5	10		1	8
61-70	10	64.5	20		2	20
71-80	4	74.5	30		3	12

$$\sum f = 40$$

$$\sum fd' = 24$$

$$(\bar{x}) = 44.5 + \frac{24}{40} \times 10 = 51.5$$

Measures of Central Tendency:-

I Median:

Test Yourself 9.2

1-soln

x	f	cf
3	4	4
4	7	11
5	6	17
6	9	26
7	10	36
8	5	41

$$N = 41$$

$$Me = \frac{N}{2} = \frac{41}{2} = 20.5$$

$$\text{Median} = 6$$

2-soln

x	f	cf
7	6	6
12	12	18
17	9	27
22	8	35
27	10	45
31	7	52
37	8	60

$$N = 60$$

$$Me = \frac{60}{2} = 30$$

$$\text{Median} = 27$$

Test Yourself 9.3:

1-solu

x	f	cf
5-10	4	4
10-15	10	14
15-20	18	32
20-25	35	67
25-30	17	94
30-35	8	102

$$\frac{N}{2} = \frac{102}{2} = 51$$

Median class = 20-25

$$L = 20$$

$$i = 5$$

$$f = 35$$

$$cf = 32$$

$$\begin{aligned} \text{Median} &= 20 + \frac{51 - 32}{35 - 32} \times 5 \\ &= 20 + \frac{19}{3} = 22 \end{aligned}$$

2-solu

x	f	cf
30-50	5	5
50-70	14	19
70-90	19	38
90-110	35	73
110-130	11	84
130-150	6	90

$$\frac{N}{2} = \frac{90}{2} = 45$$

Median class = 90-110

$$L = 90$$

$$i = 20$$

$$f = 35$$

$$cf = 38$$

$$\begin{aligned} \text{Median} &= 90 + \frac{45 - 38}{35 - 38} \times 20 \\ &= 90 + \frac{7}{-3} \times 20 \\ &= 94 \end{aligned}$$

Revision Exercise

7-5-

xi	f	c.f.
10-20	30	30
20-30	21	51
30-40	58	109
40-50	51	160

$$N = 160$$

$$Me = \frac{160}{2} = 80$$

$$\text{Median class} = 30-40$$

$$l_1 = 30$$

$$f = 58$$

$$\text{c.f.} = 51$$

$$i = 10$$

$$\text{Median} = 30 + \frac{80 - 51}{58} \times 10$$

$$= 30 + \frac{29}{58} \times 10$$

$$= 35$$

11-Sol-
Class

	xi	f	c.f.
35-40	37.5	30	30
40-45	42.5	20	50
45-50	47.5	15	65
50-55	52.5	13	78
55-60	57.5	22	100

$$Me = \frac{100}{2} = 50$$

$$\text{Median class} = 45-50$$

$$l_1 = 45$$

$$f = 15$$

$$\text{c.f.} = 50$$

$$Me = l_1 + \frac{\frac{N}{2} - \text{c.f.}}{f} \times i$$

$$45 + \frac{50 - 50}{15} \times 5$$

$$\text{Median} = 47.5 + \frac{50.5 - 50}{15} \times 5$$

$$= 47.5 + \frac{0.5}{3}$$

$$= 47.5 + \frac{5}{30}$$

$$= \frac{1425 + 5}{30} = \frac{1430}{30} = 47.66$$

Mean

I Continuous Series

10-Sub

x	m	f	fm
0-4	2	7	14
4-8	6	9	54
8-12	10	16	160
12-16	14	8	112
16-20	18	6	108
20-24	22	4	88

$$\Sigma f = 40 \quad \Sigma fm = 536$$

$$\bar{x} = \frac{\Sigma fm}{\Sigma f} = \frac{536}{40} = \frac{268}{20} = \frac{134}{10} = 13.4$$

13-Sub

x	f	m	d = x - A	d' = $\frac{d}{C}$	fd'
100-200	10	150	-200	-2	-20
200-300	18	250	-100	-1	-18
300-400	12	350 = A	0	0	0
400-500	20	450	100	1	20
500-600	40	550	200	2	80

$$\Sigma f = 100$$

$$\Sigma fd' = 62$$

fm	1500	4500	4200	9000	22000
fd	-2000	-1800	0	2000	8000

$$\Sigma fd = 4100$$

$$\Sigma fm = 41200$$

$$\text{Direct Method} = \frac{\Sigma fm}{\Sigma f}$$

$$= 412$$

$$\text{Short-cut} = A + \frac{\Sigma fd}{\Sigma f}$$

$$= 412$$

$$\text{Step-Deviation} = A + \frac{\Sigma fd'}{\Sigma f} \times C$$

$$= 412$$

II Combined Mean

17-Sol. Total students = 50

$$x_T = 62 \text{ Kg}$$

$$x_1 = 70 \text{ Kg}$$

$$x_2 = 50 \text{ Kg}$$

Formula:

$$\bar{x}_T = \frac{N_1(x_1) + N_2(x_2)}{N_1 + N_2}$$

$$62 = \frac{N_1(70) + N_2(50)}{50}$$

$$3100 = 70N_1 + 50N_2$$

Let $N_1 = x$ and $N_2 = 50 - x$

$$3100 = 70x + 50(50 - x)$$

$$3100 = 70x + 2500 - 50x$$

$$3100 = 20x + 2500$$

$$600 = 20x$$

$$x = \frac{600}{20} = 30$$

$$N_2 = 50 - 30 = 20$$

III Corrected Mean

29-Sol. $\bar{x}(\text{wrong}) = 40$
 $N = 100$

Incorrect value = $30 + 27 = 57$

Correct value = $3 + 72 = 75$

$$\bar{x}(\text{wrong}) = 40 = \frac{\sum x}{100}$$

$$\sum x(\text{wrong}) = 4000$$

$$\therefore \text{Correct } \bar{x} = \frac{4000 - 57 + 75}{100}$$

$$\bar{x} = \frac{4000 + 18}{100}$$

$$= \frac{4018}{100} = 40.18 \text{ Kg}$$

IV Weighted Mean

x_i	w_i	$x_i w_i$
62	1	62
77	2	154
65	3	325
62	4	248
57	3	171

$$\sum x_i w_i = 960$$

$$\sum w_i = 15$$

$$\bar{x} = \frac{960}{15} = 64$$

$$\frac{2574}{15} = 171.6$$

Median

Ex-12-51

x	f	cf
30-40	20	20
40-50	5	20+5
50-60	30	50+5
60-70	5	50+5+5
70-80	32	81+5+5

$$\Sigma f = 160 = 81 + 5 + 5$$

$$\Rightarrow 78 = 5 + 5$$

$$M = 56$$

$$N = 160$$

$$\frac{N}{2} = 80$$

$$2$$

$$M = l + \frac{\frac{N}{2} - cf}{f} \times i$$

$$l = 50$$

$$\text{Median class} = 50-60$$

$$f = 30$$

$$i = 10$$

$$cf = 20 + 5$$

$$\therefore 56 = 50 + \frac{80 - (20 + 5)}{30} \times 10$$

$$56 = 50 + \frac{80 - 20 + 5}{3} = 50 + \frac{60 + 5}{3}$$

$$168 = 110 + 5$$

$$5 = 58$$

$$78 = 58 + 5$$

$$5 = 20$$

Ex-16-51

x	f	cf
0-10	12	12
10-20	14	26
20-30	8	26+8
30-40	8	34+8
40-50	16	50+8
50-60	10	60+8

$$N = 60 + 8 = 68$$

$$\text{Median} = 29$$

$$\text{Median class} = 20-30$$

$$l = 20$$

$$29 = 20 + \frac{60 + 8 - 26}{8} \times 10$$

$$f = 8$$

$$i = 10$$

$$cf = 26$$

$$18 = (60 - 52 + 8) \times 10$$

$$8 = 80$$

$$8 = 10$$

$$29 = 20 + \frac{60 + 8 - 26}{8} \times 10$$

$$9 = \frac{34 + 8}{8} \times 10$$

$$9 = \frac{240 + 108}{8}$$

$$29 = 20 + \frac{60 + 8 - 26}{8} \times 10$$

$$9 = \frac{60 + 8 - 26}{8} \times 10$$

$$18 = \frac{600 + 108 - 260}{8}$$

$$8 = \frac{340}{8}$$

$$8 = 45$$

Revision Exercise

15-sol

x	f	cf
0-10	8	8
10-20	12	20
20-30	16	36
30-40	10	46
40-50	5	51
50-60	12	63

$$\Sigma f = 64$$

$$\text{Median} = 34$$

$$\text{Median class} = 20-30$$

$$L = 20$$

$$i = 10$$

$$E = 16$$

$$fcf = 20$$

$$\therefore 34 = 20 + \frac{64 - 20}{16} \times 10$$

$$14 = \frac{44 + f}{16} \times 10$$

$$14 \times 16 = 44 + f \times 10$$

$$f = 16$$

17-sol

x	f	cf
0-10	10	10
10-20	20	30
20-30	5	35
30-40	40	75
40-50	5	80
50-60	25	105
60-70	15	120

$$\Sigma f = 110 + 5 + 5 = 120$$

$$\therefore 5 + 5 = 60$$

(1)

$$\text{Median} = 35$$

$$\text{Median class} = 20-30$$

$$L = 20$$

$$i = 10$$

$$f = 5$$

$$cf = 30$$

$$\therefore 35 = 20 + \frac{120 - 30}{5} \times 10$$

$$15 = \frac{(85 - 30) \times 10}{5}$$

$$15 = \frac{550}{5}$$

$$5 = \frac{550}{11} = 50$$

$$\text{Putting } 5 = 50 \text{ in (2)}$$

$$25 + f = 60$$

$$f = 35$$

Quartile

Test Yourself 9.6

1-Sol.

x	f
1	39
2	45
3	50
4	56
5	62
6	68
7	73
8	77
9	84
10	88
11	91

$$Q_1 = \frac{11+1}{4} = 3$$

50

$$Q_2 = 2 \times \frac{11+1}{4} = 3 \times 3 = 9$$

$$= 84$$

2-Sol.

x	f	cf
12	4	4
28	6	10
33	9	19
47	10	29
55	7	36
62	3	41
75	1	42

$$N = \sum f = 42$$

$$Q_1 = \frac{42+1}{4} = \frac{43}{4} = 10.75$$

$\therefore 10.75$ falls in cf. 19 and therefore

$$Q_1 = 33$$

$$Q_3 = 3 \times \frac{42+1}{4} = \frac{3 \times 43}{4} = \frac{129}{4} = 32.25$$

$\therefore 32.25$ falls under the cf 36 and therefore

$$Q_3 = 55$$

3-50

x	f	cf
0-10	11	11
10-20	12	23
20-30	19	42
30-40	20	62
40-50	18	80

$$N = \sum f = 80$$

$$Q_1 = \frac{80+1}{4} = \frac{81}{4} = 20.25$$

$\therefore 20.25$ lies in cf 23.

$$\therefore l = 10$$

$$cf = 11$$

$$f = 12$$

$$i = 10$$

$$\therefore Q_1 = 10 + \frac{20.25 - 11}{12} \times 10$$

$$= 10 + \frac{9.25}{12} \times 10$$

$$= 10 + \frac{925}{120} \times 10 = \frac{1200 + 925}{120} = \frac{2125}{120}$$

$$= \frac{425}{24} = 17.7$$

$$Q_3 = 3 \times \frac{81}{4} = \frac{243}{4} = 60.75$$

$\therefore 60.75$ lies in cf 62

$$\therefore l = 30$$

$$i = 10$$

$$cf = 42$$

$$f = 20$$

$$\therefore Q_3 = 30 + \frac{60.75 - 42}{20} \times 10$$

$$= 30 + \frac{18.75}{2}$$

$$= 30 + 9.37$$

$$= 39.37$$

Revision Exercise

20-Sol	x	f	cf
	58	2	2
	59	3	5
	60	6	11
	61	15	26
	62	10	36
	63	5	41
	64	4	45
	65	3	48
	66	1	49

$$N = \Sigma f = 49$$

$$Q_1 = \frac{49+1}{4} = \frac{50}{4} = 12.5$$

$\therefore 12.5$ lies in cf 26.

$$L = 61$$

$$i = 1$$

$$cf = 11$$

$$f = 15$$

$$\therefore Q_1 = 61 + \frac{12.5 - 11}{15} \times 1$$

$$= 61 + \frac{1.5}{15} = 61 + \frac{1}{10} = \frac{610+1}{10} = 61.1$$

$$Q_3 = \frac{49+1}{4} \times 3 = \frac{150}{4} = 37.5$$

$\therefore 37.5$ lies in cf 41.

$$L = 63$$

$$i = 1$$

$$f = 5$$

$$cf = 36$$

$$\therefore Q_3 = 63 + \frac{37.5 - 36}{5} \times 1$$

$$= 63 + \frac{1.5}{5} = 63.3$$

$$\text{Median} = \frac{N}{2} = \frac{49}{2} = 24.5$$

$\therefore 24.5$ lies in cf 36.

$$L = 62$$

$$i = 1$$

$$f = 10$$

$$cf = 26$$

$$\text{Median} = 62 + \frac{24.5 - 26}{10} \times 1$$

$$= 62 + \frac{-1.5}{10} \times 1 = 62 - \frac{1.5}{10} = \frac{620 - 1.5}{10} = \frac{618.5}{10}$$

$$= 61.85$$

22. Solu

x	f	cf
0-10	10	10
10-20	8	18
20-30	12	30
30-40	16	46
40-50	14	60
50-60	6	66
60-70	18	84

$$N = \sum f = 84$$

$$Q_1 = \frac{84+1}{4} = \frac{85}{4} = 21.25$$

$\therefore 21.25$ lies in cf 30

$$l = 20$$

$$i = 10$$

$$f = 12$$

$$cf = 18$$

$$Q_1 = 20 + \frac{21.25 - 18}{12} \times 10$$

$$= 20 + \frac{3.25}{12} \times 10 = 20 + \frac{32.5}{12} = 22.8$$

$$Q_2 = 3 \times 22.8 = 68.4$$

$\therefore 68.4$ lies in cf 66

$$l = 50$$

$$i = 10$$

$$cf = 60$$

$$f = 6$$

$$Q_3 = 50 + \frac{68.4 - 60}{6} \times 10$$

$$= 50 + \frac{8.4}{6} \times 10$$

$$= 50 + \frac{84}{6}$$

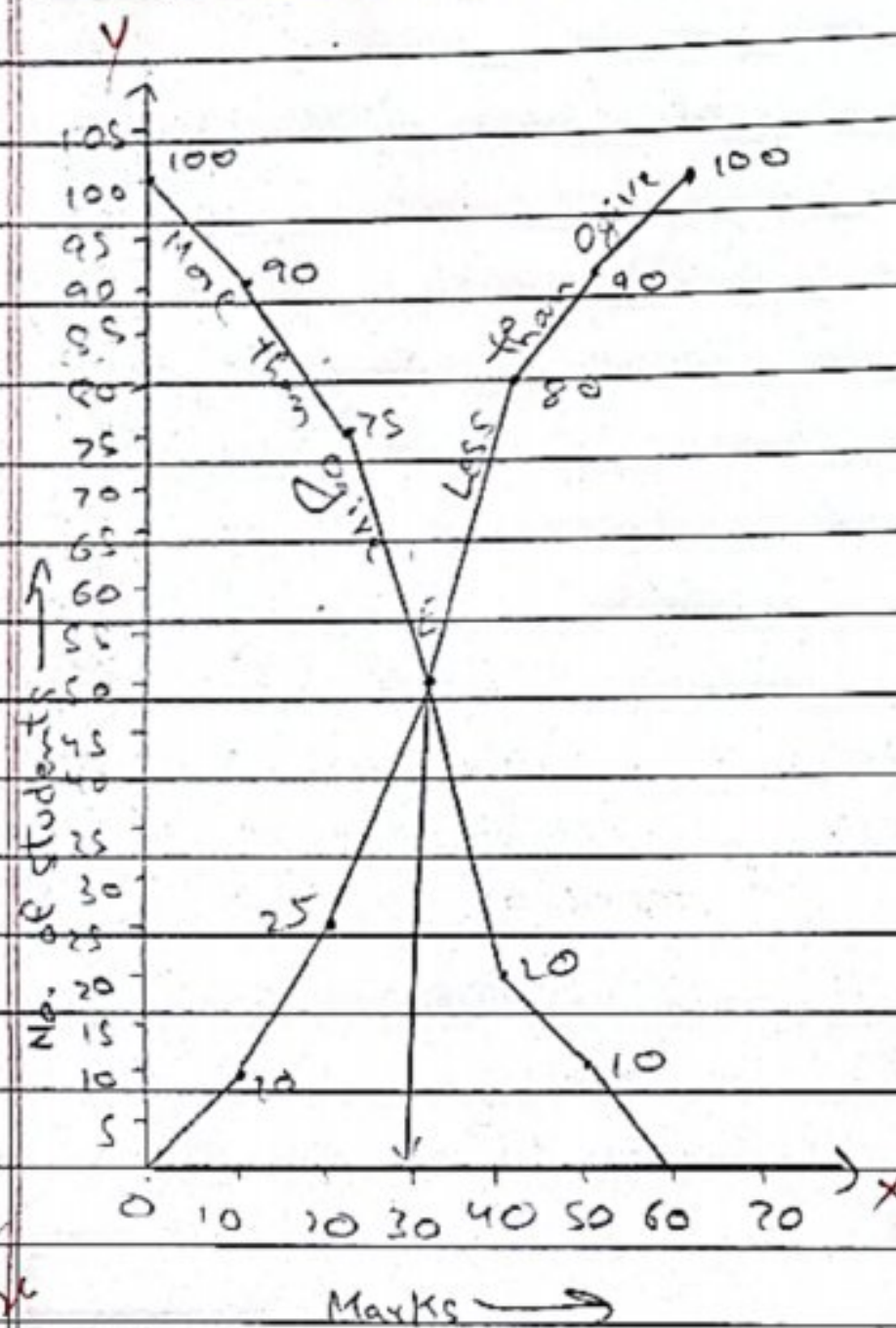
$$= 50 + 14$$

$$= 64$$

Graphic Location of Median

23-540

CI	f	Less than	More than
0-10	10	10	100
10-20	15	25	90
20-30	25	50	75
30-40	30	80	50
40-50	10	90	20
50-60	10	100	10
60-70	-	-	-
70-80	-	-	-
80-90	-	-	-
90-100	-	-	-



Median = 30

Class Test of Mean

Q1

Calculate Mean from the following Data

x	f
0-4	7
4-8	9
8-12	16
12-16	8
16-20	6
20-24	4

Solu

m	x	f	d = m - A	fd
2	0-4	7	-8	-56
6	4-8	9	-4	-36
10 = A	8-12	16	0	0
14	12-16	8	4	32
18	16-20	6	8	48
22	20-24	4	12	48
		$\Sigma f = 50$		$\Sigma fd = 36$

Short cut Method:

$$(\bar{x}) = 10 + \frac{36}{50} = \frac{500 + 36}{50} = \frac{536}{50} = 10.72$$

Ans 10.72

Q2 The average height of a group of 40 students was calculated as 155 cm. It was later discovered that the height of 1 student was read as 157 instead of 137 cm. Calculate the correct average height.

Soln Wrong $(\bar{x}) = 155$
Total students = 40

$$\therefore 155 = \frac{\sum fx}{40}$$

$$(\text{Wrong}) \sum fx = 6200$$

Incorrect value = 157

Correct value = 137

$$(\bar{x})_{\text{Correct}} = \frac{6200 - 157 + 137}{40}$$

$$= \frac{6200 - 20}{40} = \frac{6180}{40} = 154.5 \text{ cm}$$

Ans 154.5 cm

Q3 Locate the missing frequency if the mean of the series is 40

x	f
20	8
30	?
40	12
50	6
60	7
70	2

Soln

$$\sum f = 35 + f$$

$$\sum fx = 1500 + 30f$$

$$(\bar{x}) = 40$$

x	f	fx
20	8	160
30	f	30f
40	12	480
50	6	300
60	7	420
70	2	140

Direct Method:

$$40 = \frac{1500 + 30f}{35 + f}$$

$$(35 + f) 40 = 1500 + 30f$$

$$1400 + 40f = 1500 + 30f$$

$$10f = 100$$

$$f = 10$$

Ans $f = 10$

21/4/20

Class Test of Median & Mode

Q1 Calculate mode from the following data

Marks	f
0-10	5
10-20	12
20-40	40
40-50	32
50-70	18

Solu

Modal class = 20-40

$$f_0 = 12$$

$$f_1 = 40$$

$$f_2 = 32$$

$$x = 10$$

$$l = 20$$

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times x$$

$$= 20 + \frac{40 - 12}{80 - 12 - 32} \times 10$$

$$= 20 + \frac{28 \times 10}{37}$$

$$= 20 + \frac{280}{37}$$

$$= 20 + 7.56$$

$$= 27.56$$

Q2

Calculate median from the following data

Size	f
10-20	30
20-30	21
30-40	58
40-50	51

Solu

$$N = \frac{N+1}{2}$$

$$= \frac{161}{2} = 80.5$$

$\therefore 80.5$ lies in
of 109

$\therefore$ Median class = 30-40

$$l = 30$$

$$x = 10$$

$$f = 58$$

$$cf = 51$$

$$\text{Median} = 30 + \frac{80.5 - 51}{58} \times 10$$

$$= 30 + \frac{29.5}{58} \times 10 = 30 + \frac{295}{58}$$

$$= 30 + 5 = 35$$

x	f	cf
10-20	30	30
20-30	21	51
30-40	58	109
40-50	51	160

Rough work

$$\begin{array}{r} 58 \times 58 \\ \times 3 \\ \hline 174 \end{array}$$

$$\begin{array}{r} 58 \times 58 \\ \times 3 \\ \hline 174 \end{array}$$

1-Solo Highest frequency = 40

Modal class: 20-40

$$f_0 = 12$$

$$f_1 = 40$$

$$f_2 = 32$$

$$f_3 = 20$$

$$i = 20$$

$$\text{Mode} = 20 + \frac{40 - 12}{80 - 12 - 32} \times 20$$

$$= 20 + \frac{28}{36} \times 20$$

$$= 20 + \frac{560}{36}$$

$$= 20 + \frac{280}{18}$$

$$= 20 + \frac{140}{9}$$

$$= 20 + 15.5$$

$$= 35.5$$